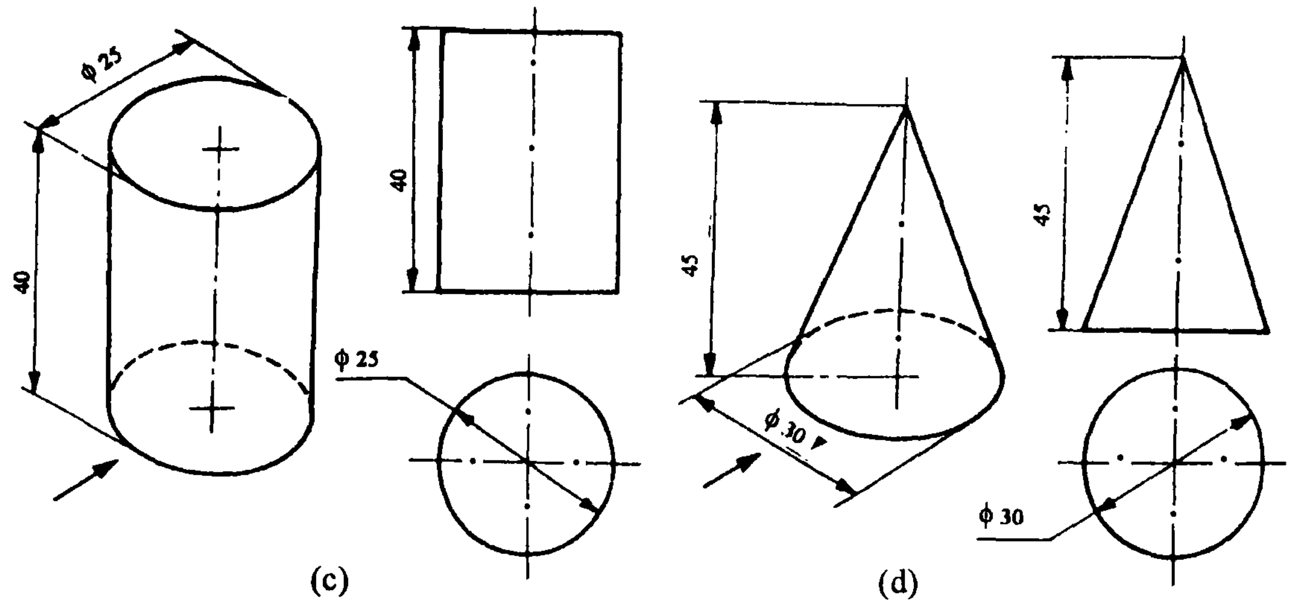


Fig. 6.46

6.9.3 Three View Drawings

In general, three views are required to describe most of the objects. In such cases the views normally selected are : the front view, top view and left or right side view. Fig.6.47 shows an example in which three views are essential to describe the object completely.

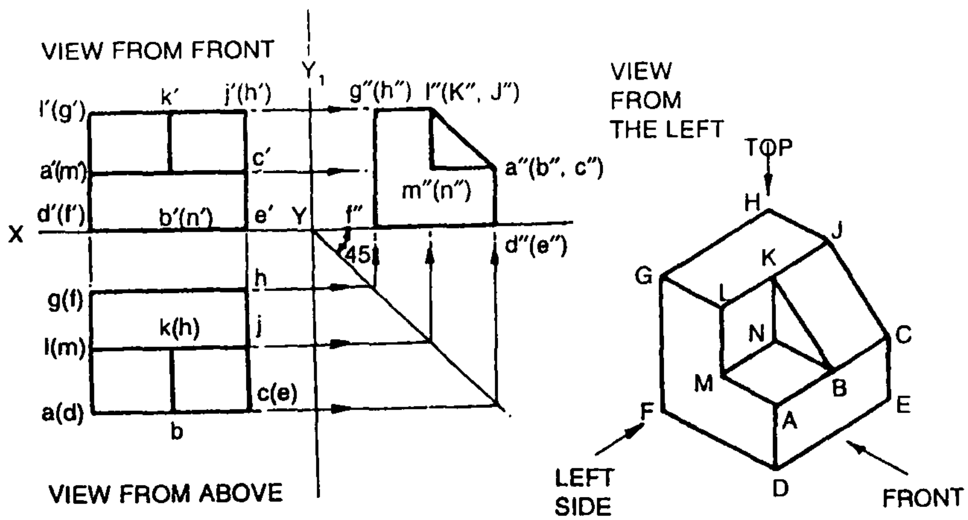


Fig. 6.47 Three View Drawing

6.9.4 Development of Missing Views

When two views of an object are given the third view may be developed by the use of mitre line as described in the following example.

(a) To develop the right side view from the given front and top views.

Construction (Fig. 6.48)

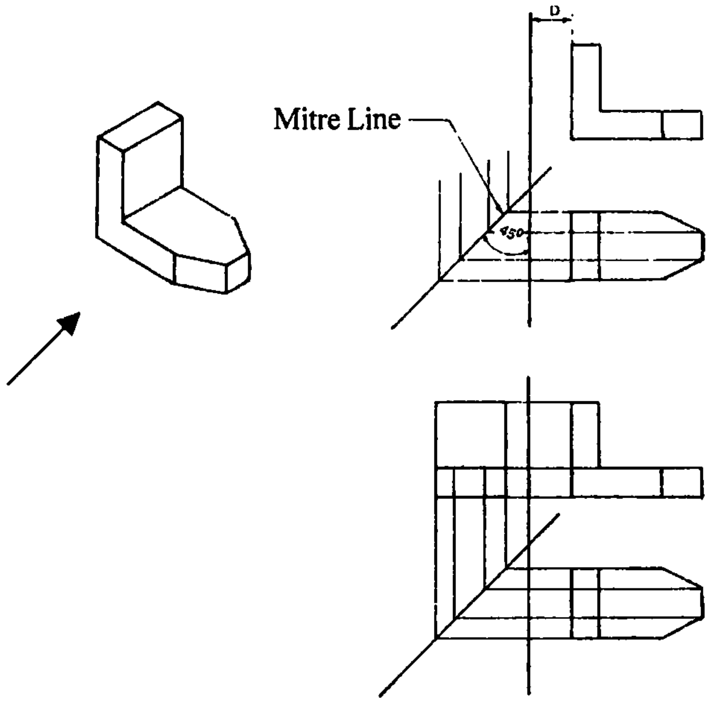


Fig. 6.48

1. Draw the given front and top views.
2. Draw projection lines to the left of the top view.
3. Draw a vertical reference line at any convement distancd D from the front view.
4. Draw a mitre line at 45° to the vertical
5. Through the points of intersection between the mitre line and the above projection lines draw vertical projection lines.
6. Join the points of intersection in the ordr and obtain the required view.

(b) Figure 5.49 illustrates the method of obtianing the top views from the given front and left side views.

(c) Figure 6.50 shows the correct positioning of the three orthographic views.

Examples

For examples given note the following:

Figure a - Isometric projection

Figure b - Orthographic projections

Direction of arrow - Direction to obtain the front view.

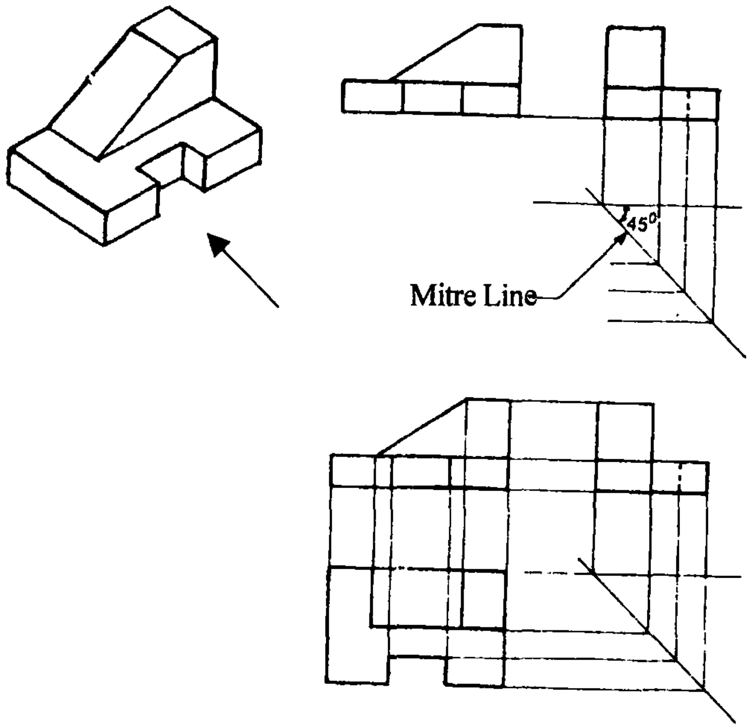


Fig. 6.49

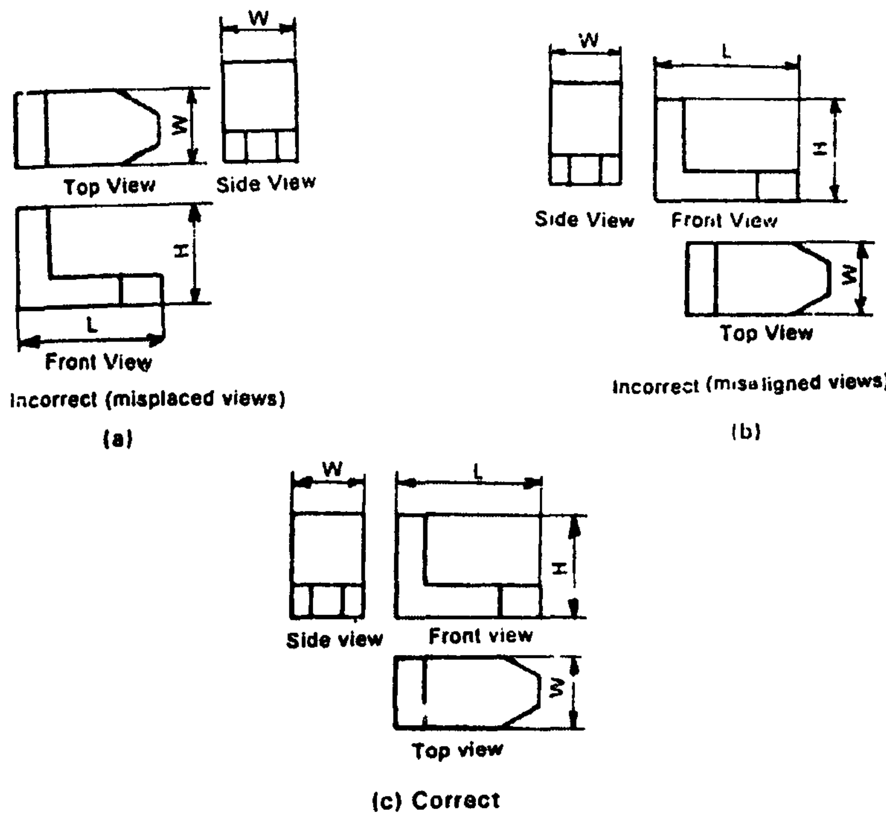


Fig. 6.50

Example 1

In the following figures form 6.51 to 6.72 the isometric projection of some solids and machine components are shown for which the three orthgraphic views are given in first angle projection.

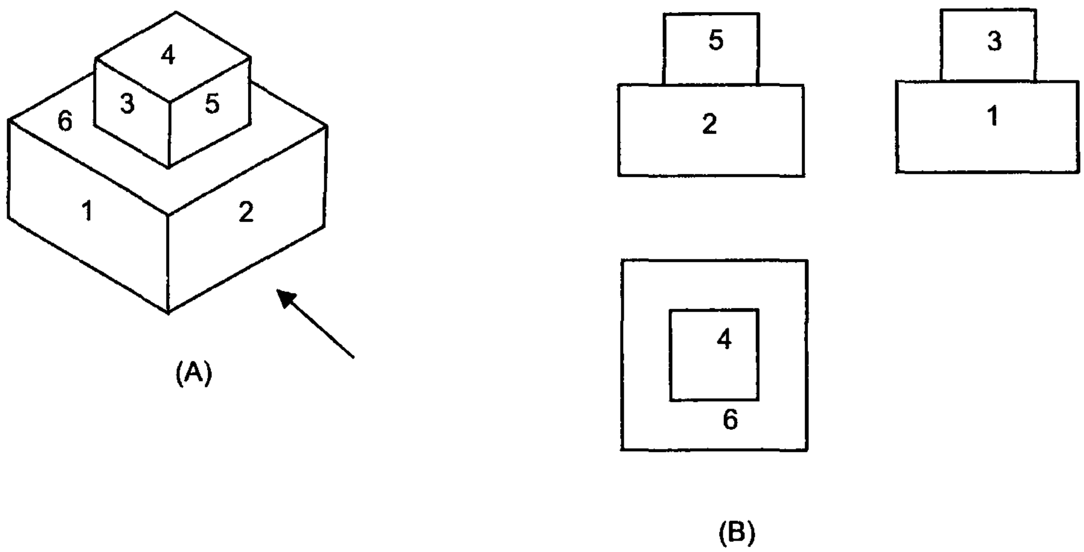


Fig. 6.51

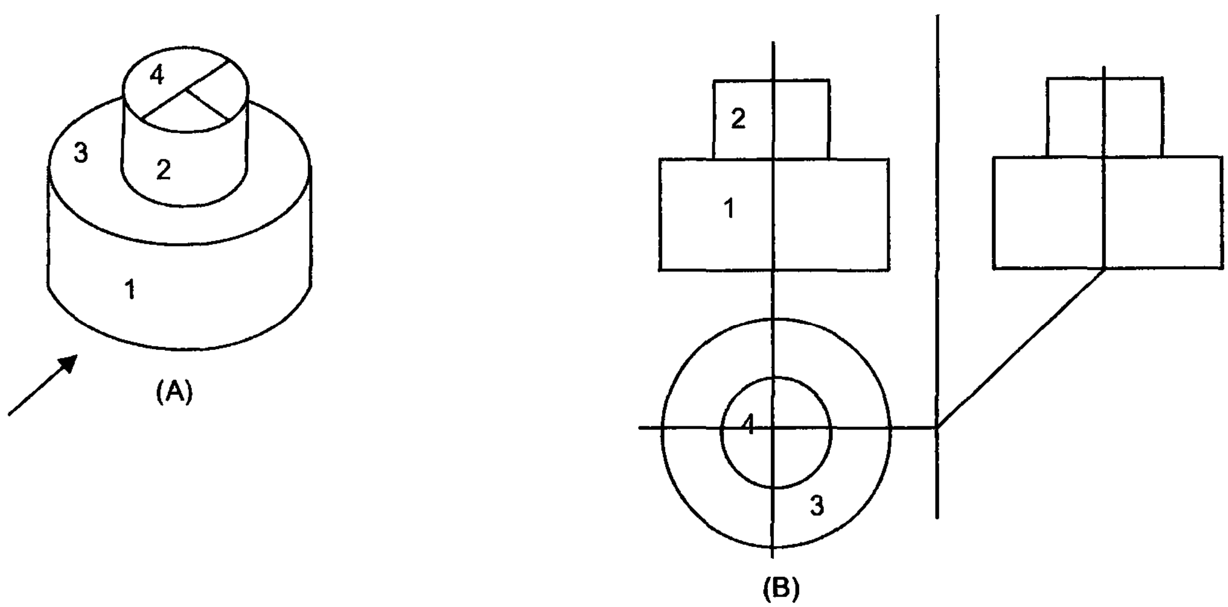


Fig. 6.52

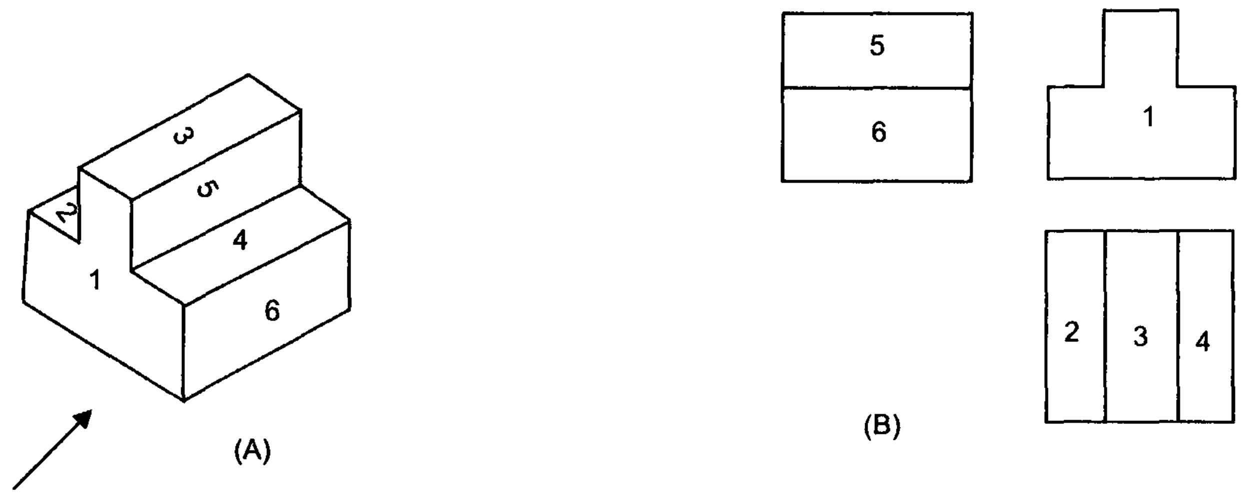


Fig. 6.53

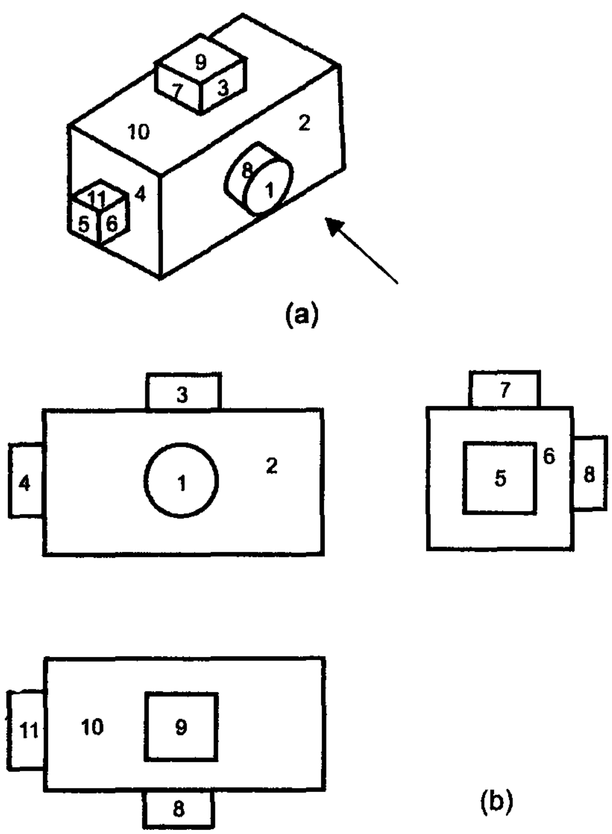


Fig. 6.54

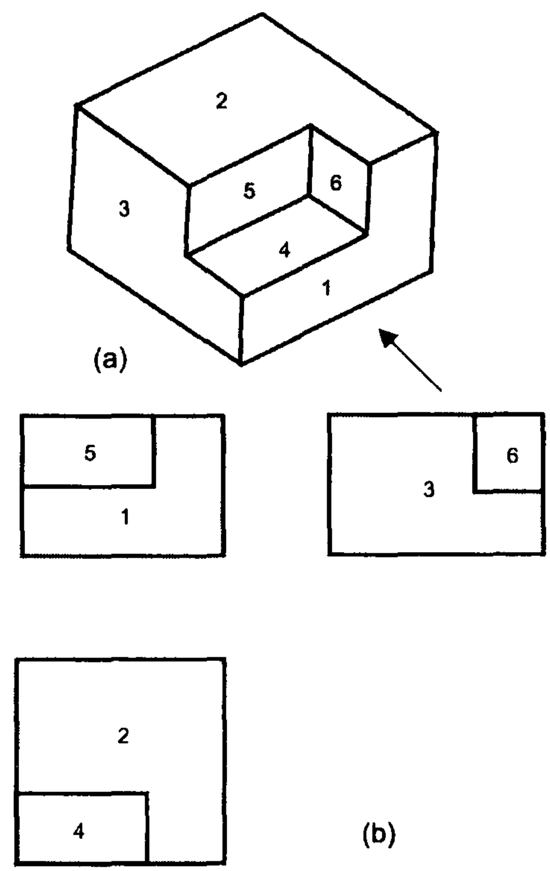
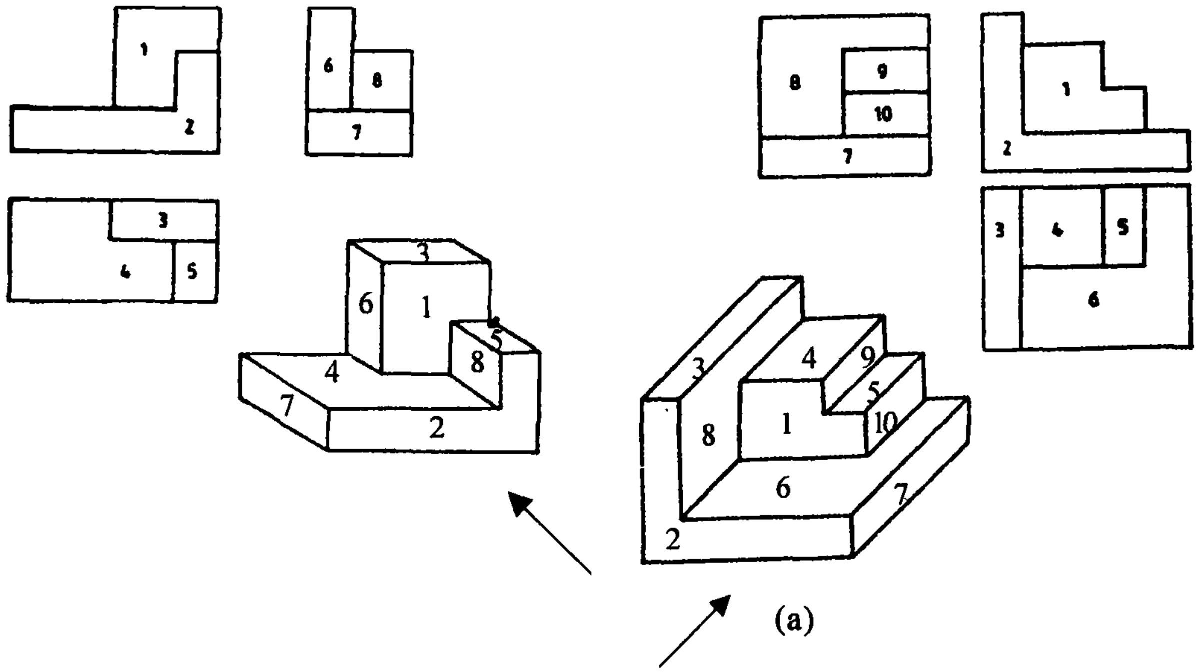
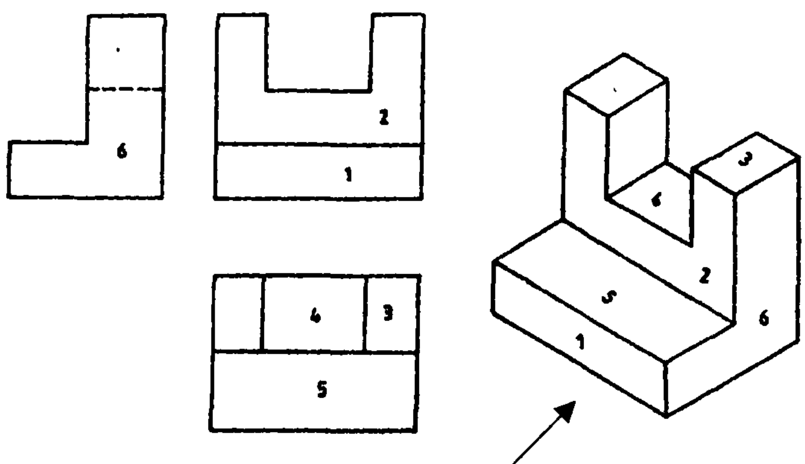


Fig. 6.55



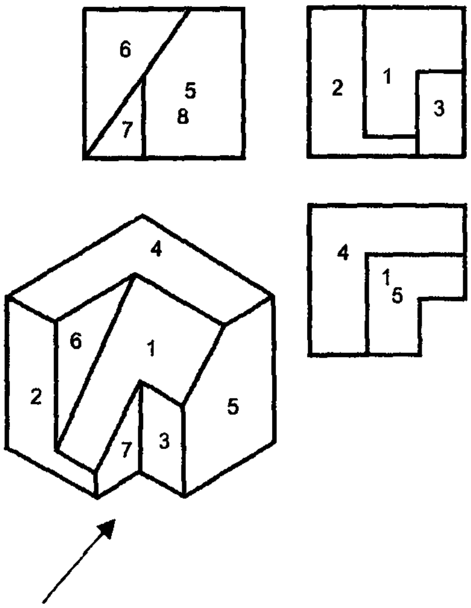
(a)

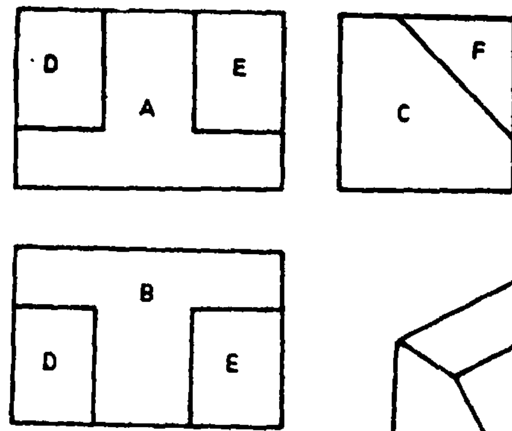
Fig. 6.56(a) & 6.57



(a)

Fig. 6.58





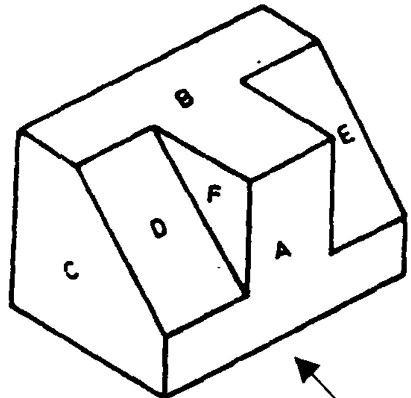
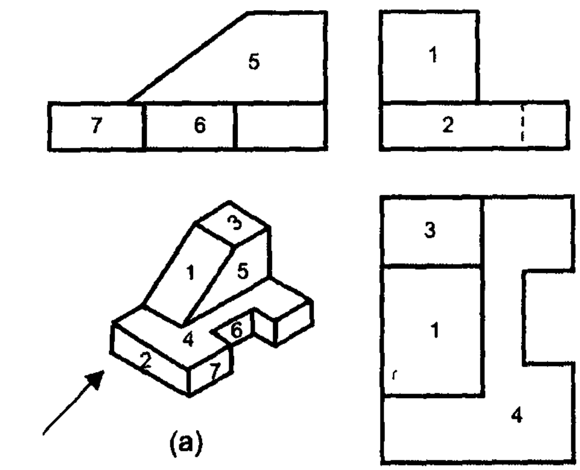


Fig. 6.59 & 6.60



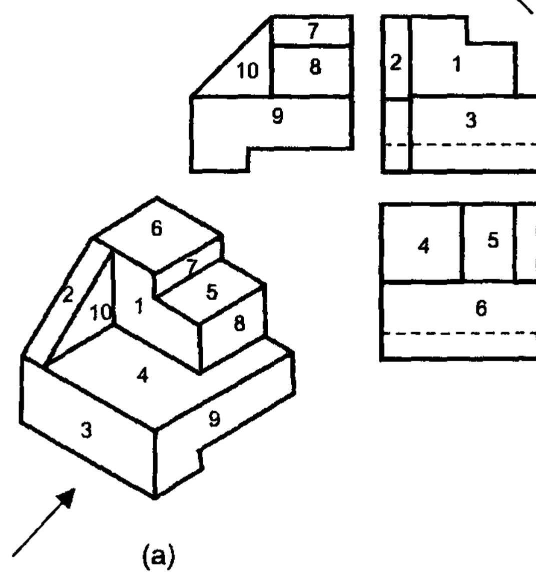


Fig. 6.61 & 6.62

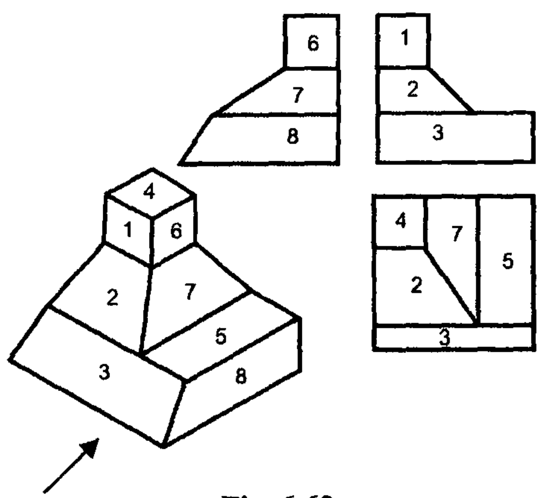


Fig. 6.63

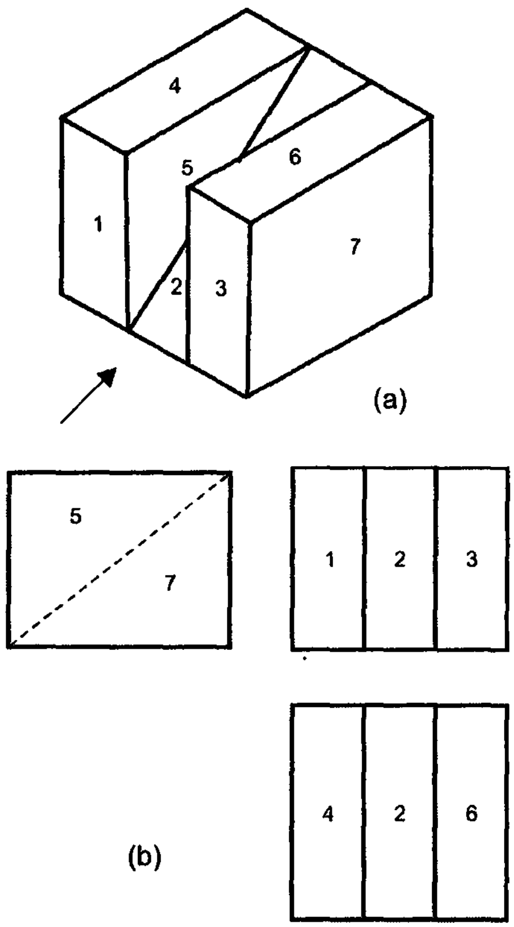


Fig. 6.64

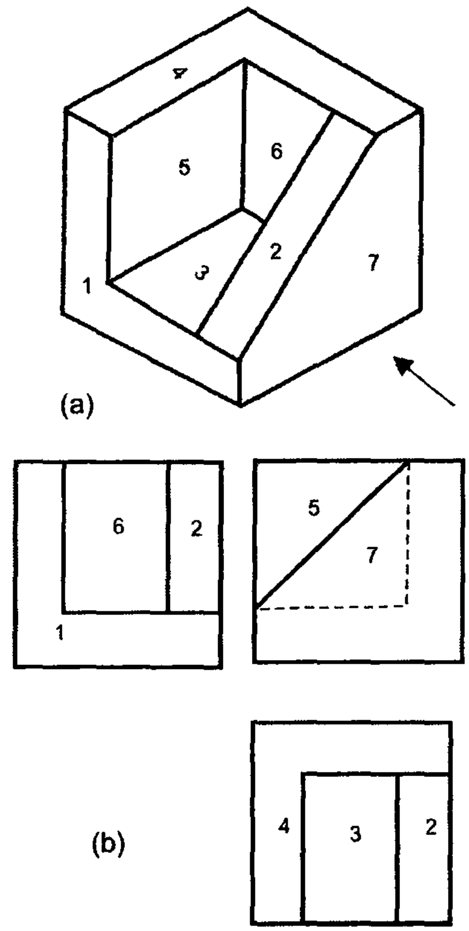


Fig. 6.65

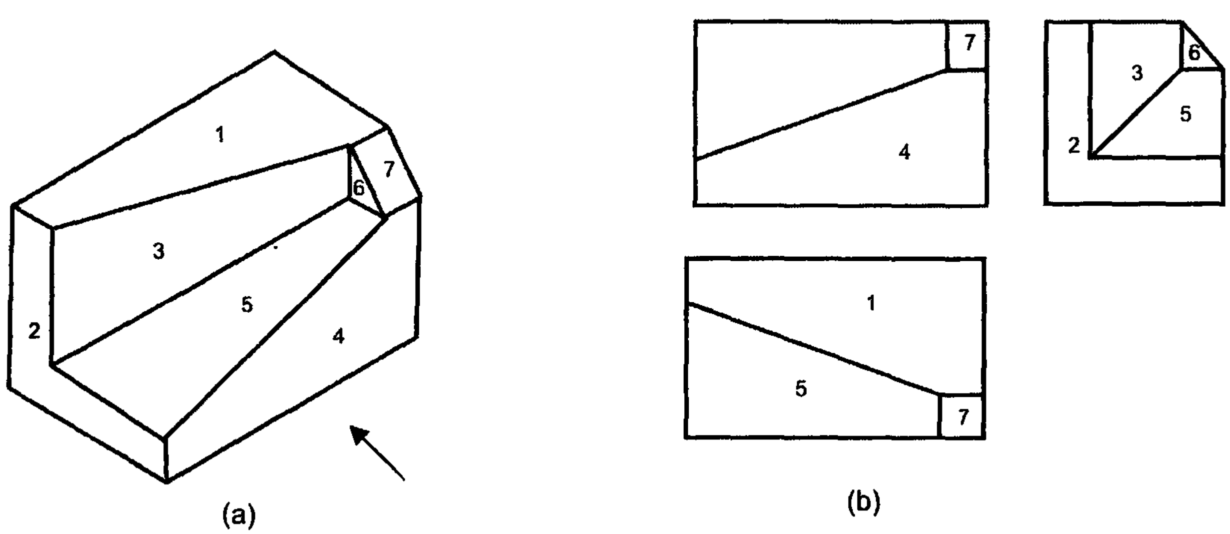
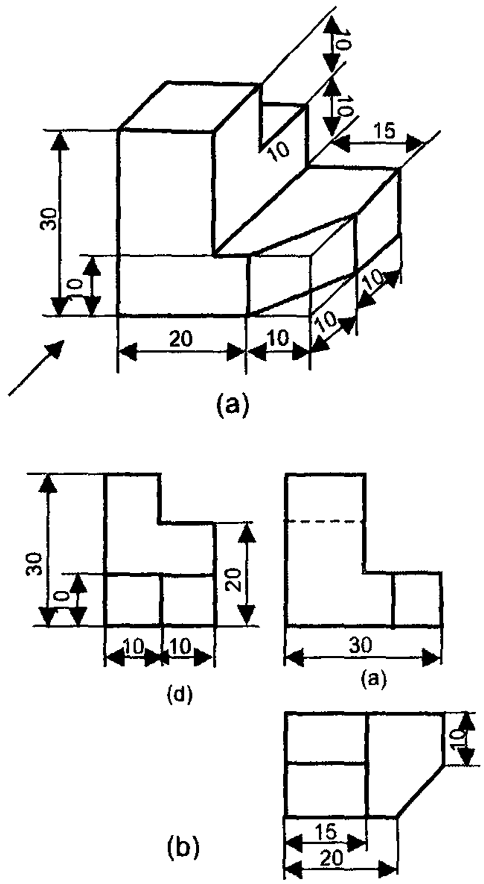


Fig. 6.66



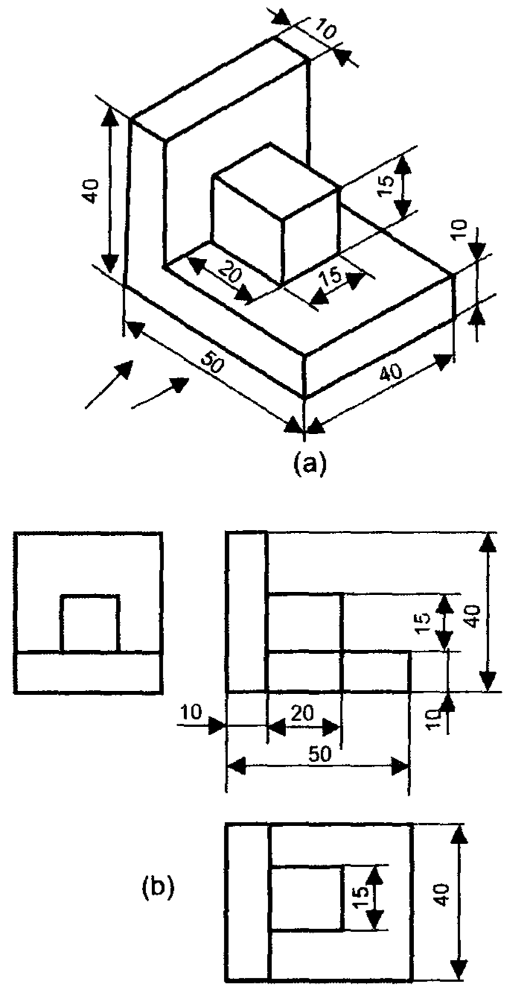


Fig. 6.67 & 6.68

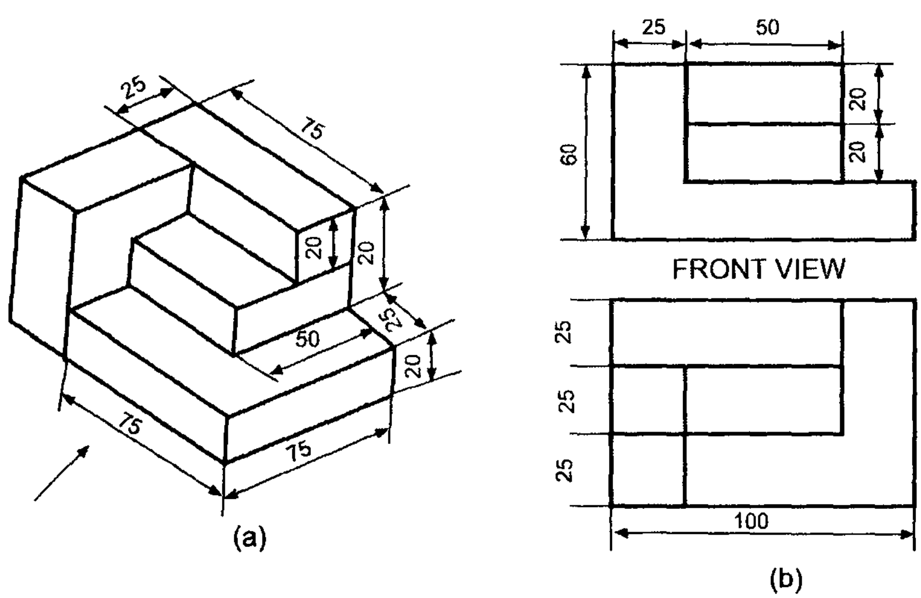


Fig. 6.69

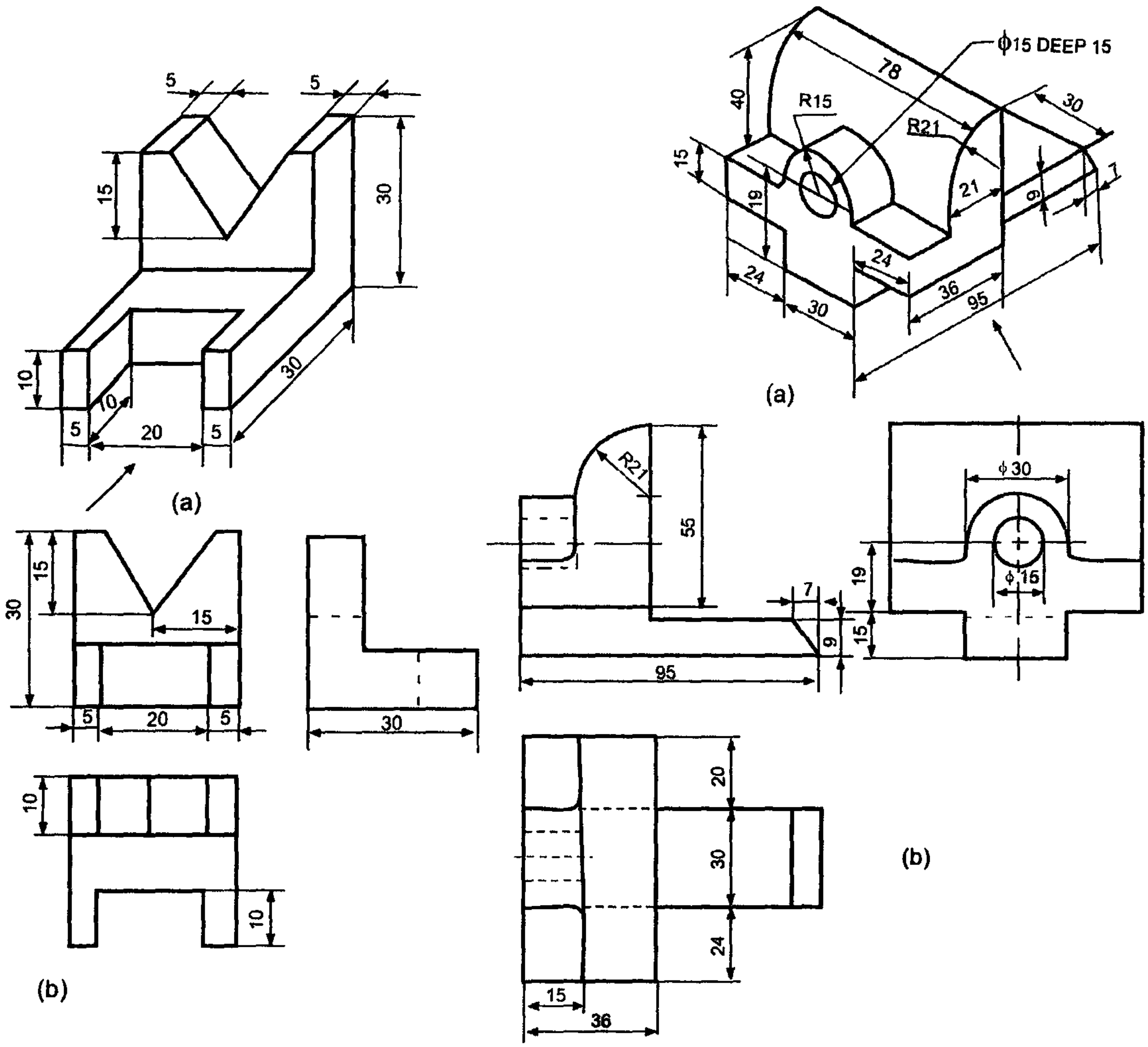

5
5
15
30
10
10
30
5
20
5
(a)
30
15
15
5
20
5
30
10
10
(b)
ϕ15 DEEP 15
78
40
R15
R21
30
15
19
21
9
7
24
24
30
36
95
(a)
R21
55
7
9
15
95
ϕ 30
19
ϕ 15
20
30
24
15
36
(b)

Fig. 6.70 & 6.71

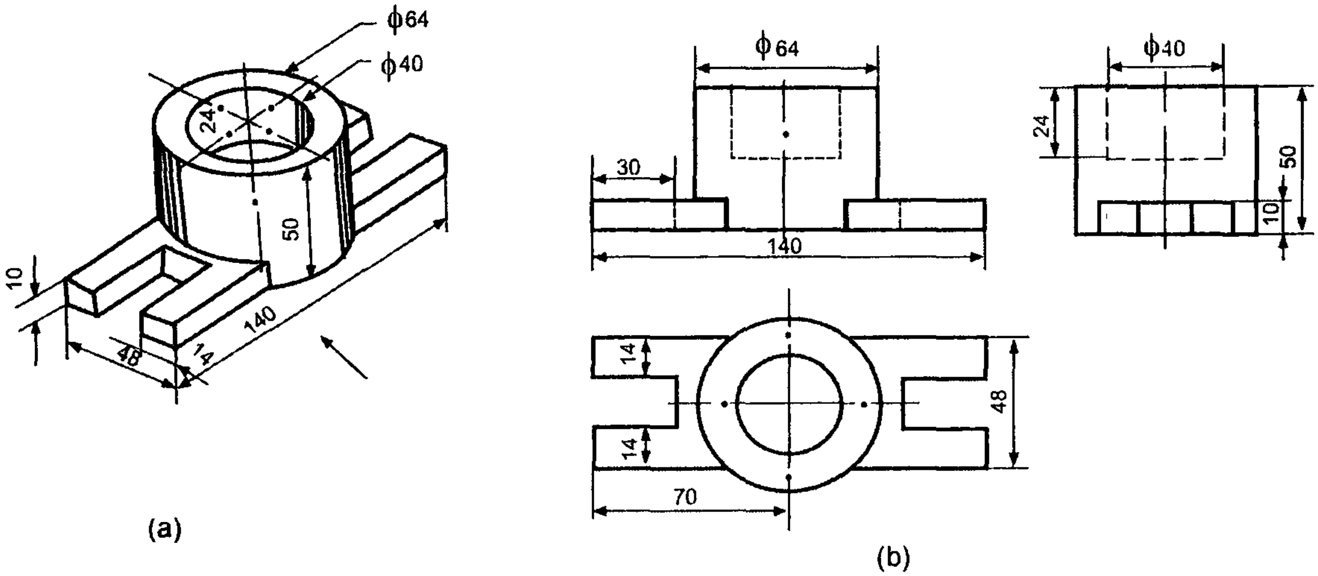


Fig. 6.72

Example 2

Figures 6.73 and 6.76 show the isometric views of cerstain objects A to H along with their orthographic views. Identify the front, top or side views of the objects and draw the third view.

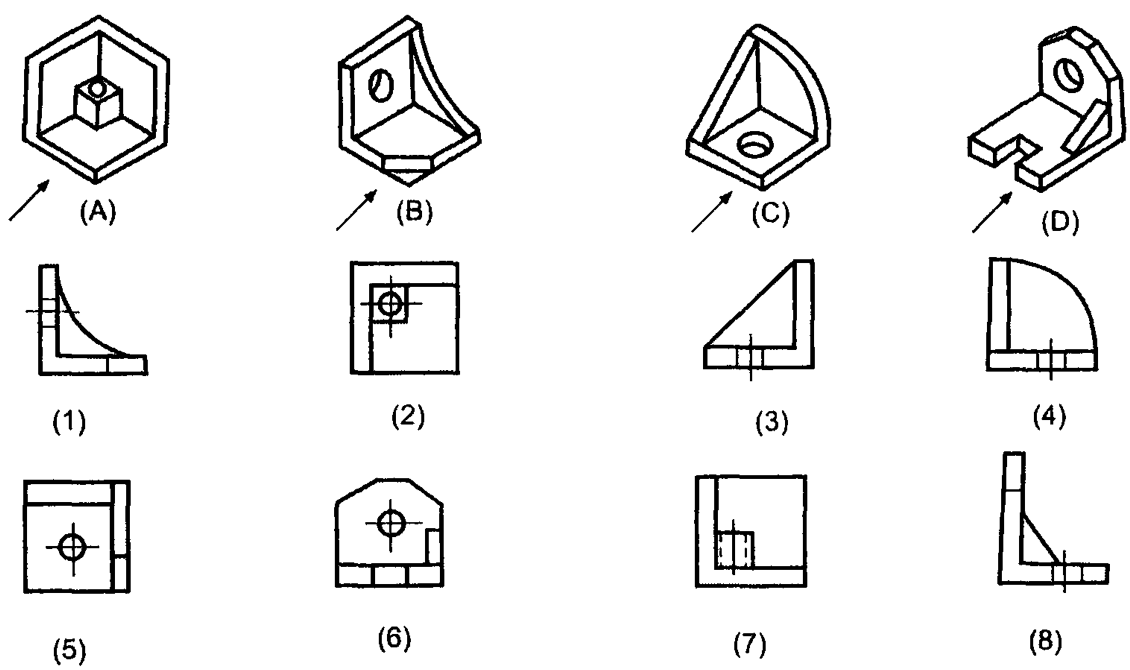


Fig. 6.73

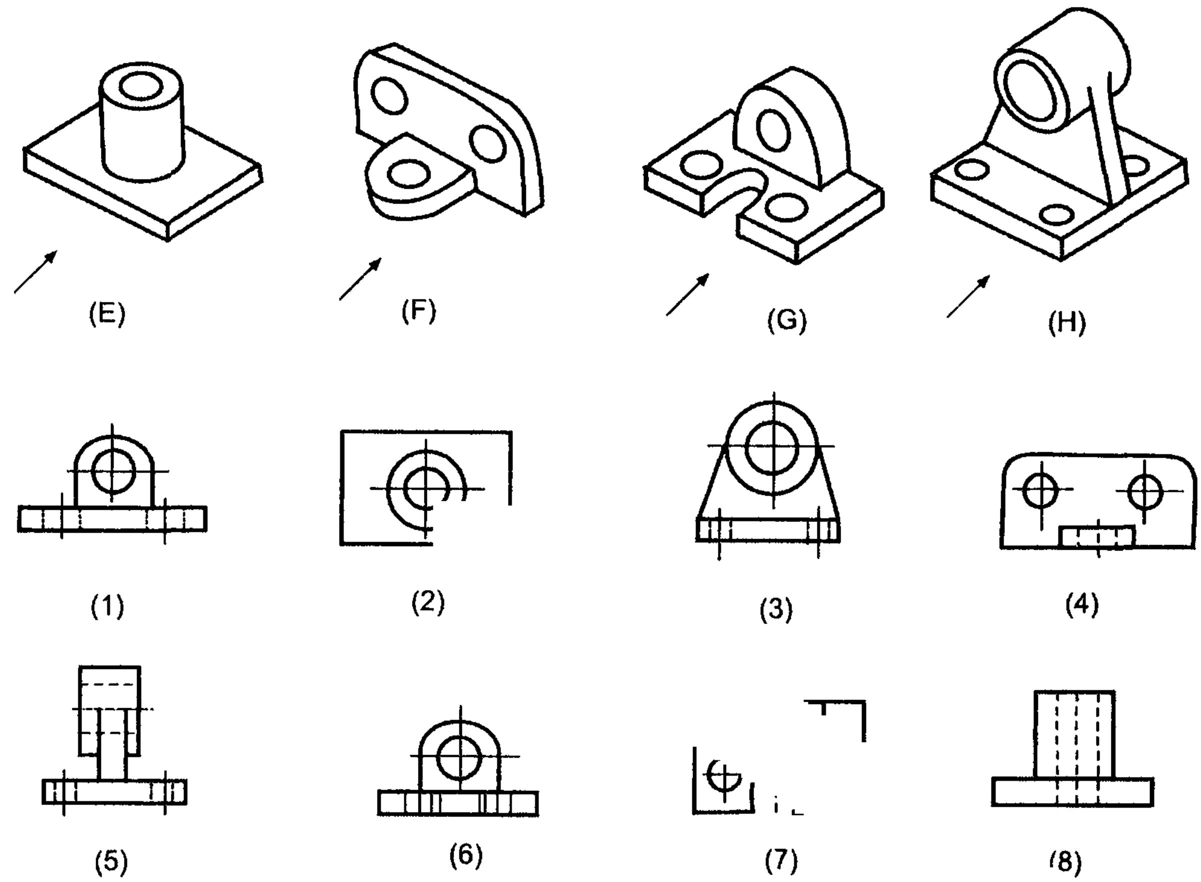
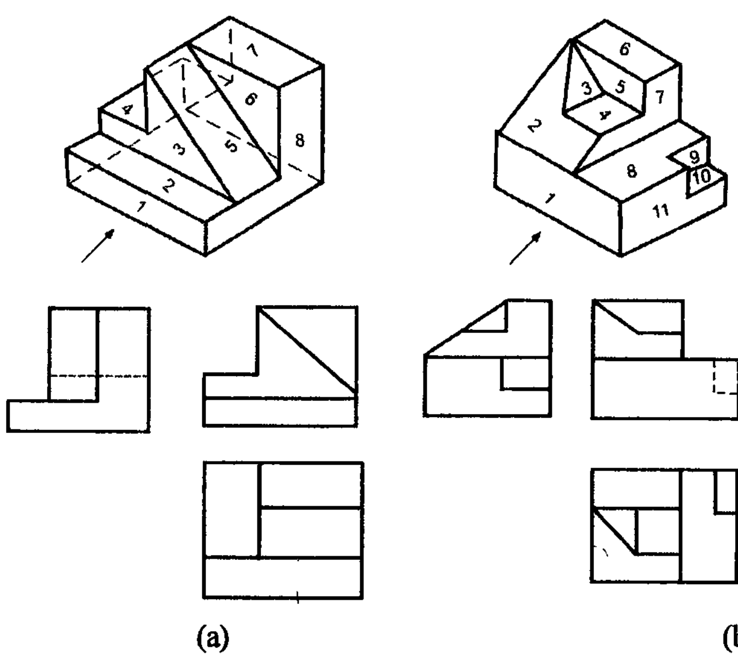
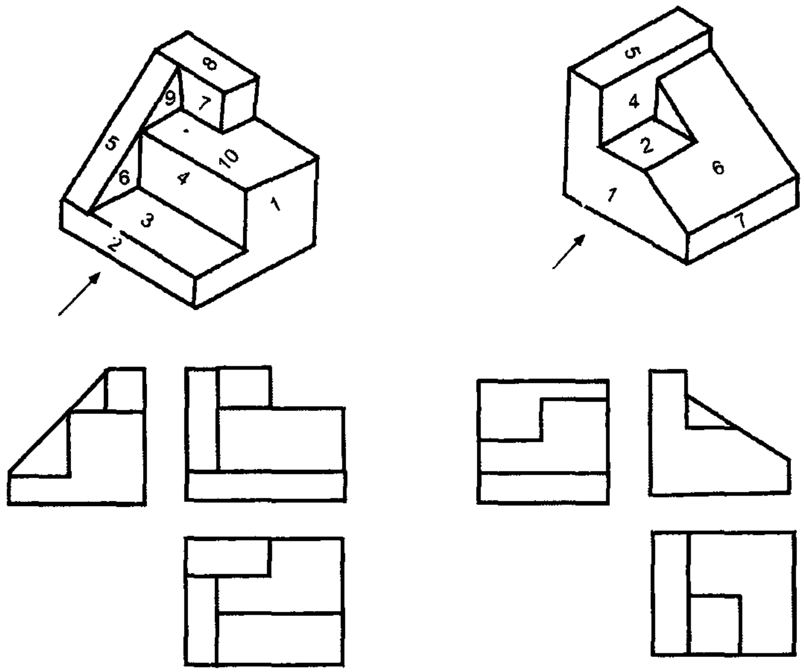


Fig. 6.74

Exercise 2

Study the isometric views in Figures 6.75 and identify the surfaces and number them looking in the direction of the arrows.



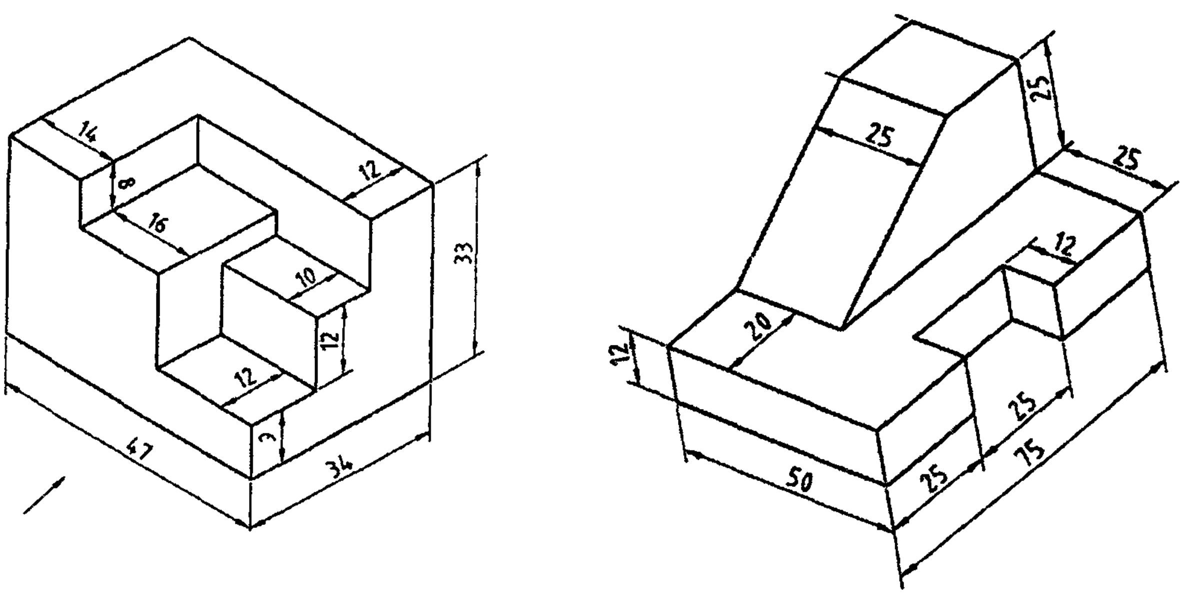


(c) (d)

Fig. 6.75

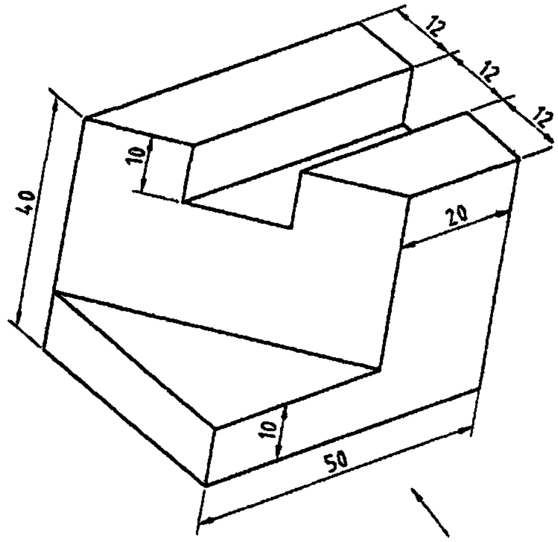
Exercise 3

Study the isometric views in Figures 6.76 and draw the orthographic views looking in the direction of the arrows and number the surfaces.

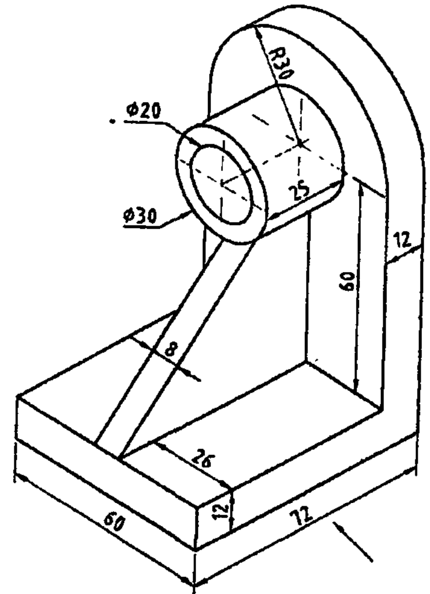


(a) (b)

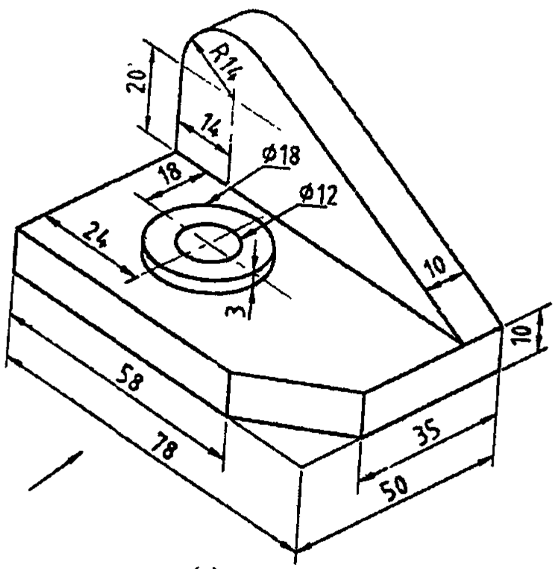
Fig. 6.76(a) &(b)



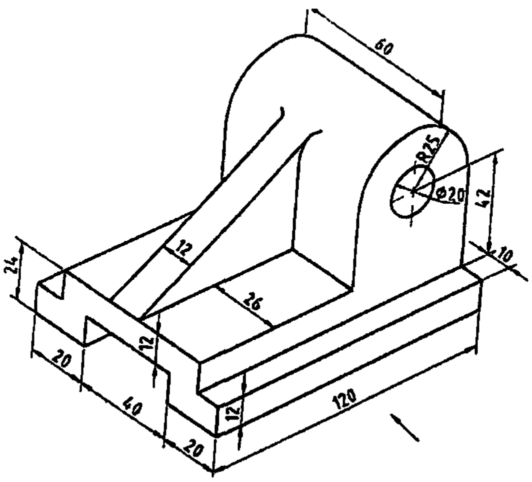
(c)



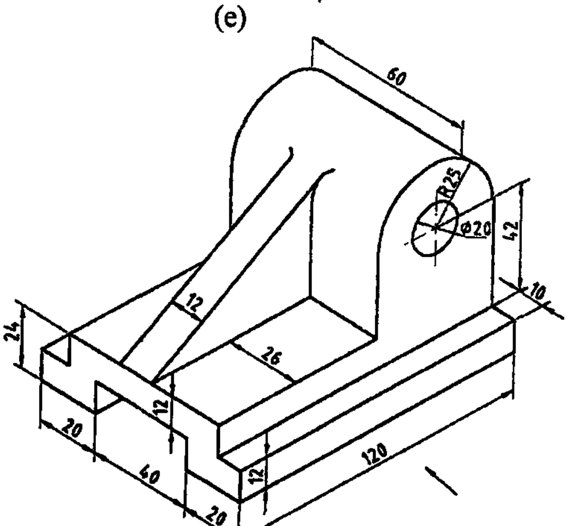
(d)



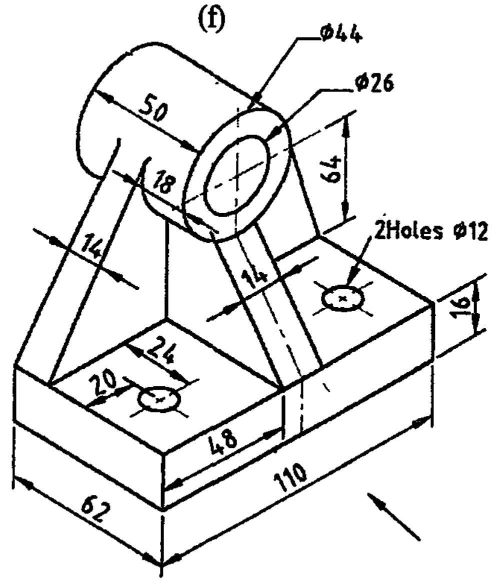
(e)



(f)



(g)



(h)

Fig. 6.76(e)(f)(g)&(b)